

Cultural ecosystem services provided by rivers across diverse social-ecological landscapes: A social media analysis

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ABSTRACT

Cultural ecosystem services (CES) are an important component of the benefits that humans derive from nature. Yet, research on CES at landscape scales has lagged behind other ecosystem services, due to the difficulty measuring CES across broad scales and the uncertainty about the mechanisms linking CES provisioning to biophysical characteristics. Social media data has emerged as an important tool for quantifying CES. We applied a bottom-up, data-driven approach to capture rich information about CES from the text (title, tags, descriptions) associated with geo-located Flickr images across river ecosystems in Idaho, USA. We address the following four questions: 1) What CES do people obtain from rivers in Idaho? 2) How does overall CES provisioning vary across ecosystem characteristics? 3) How do specific CES relate to ecosystem characteristics? and 4) Do CES bundle in ways that parallel traditional CES categories? Overall, we were able to identify diverse CES not typically examined in empirical studies as well as how landscape features support CES provisioning. Relating CES to biophysical, social, and built characteristics of the ecosystem, CES provisioning was significantly but weakly associated with access to rivers, land cover, and land ownership. The importance of social and built characteristics of the landscape suggests the cascade model in which ecosystem services flow from ecosystem structure has limited utility for specifically addressing CES. In addition, we found specific CES are more strongly associated with landscape variables than general CES categories. This difference highlights that traditional CES categories fail to reflect how individual CES are distributed over space. Text provides information-rich source for moving beyond a small number of broad CES categories and understanding not only the diversity of ways people interact with landscapes, but what landscape features support these uses.

1. Introduction

Recognition of the non-monetary benefits that river ecosystems provide to humans and the need to quantify those benefits reaches back as far as the 1960s (Leopold, 1969), well before the concept of ecosystem services was formally defined (Millennium Ecosystem Assessment, 2005). Quantifying patterns of ecosystem service supply, demand, and flow at landscape scales is critical for understanding spatial relationships between ecosystems services, landscape attributes and the management implications for ecosystem services (Bennett et al., 2009; Plieninger et al., 2015; Raudsepp-Hearne et al., 2010). However, landscape-scale research on cultural ecosystem services (CES) such as aesthetics, spiritual and religious values, education, and recreation has lagged (Casalegno et al., 2013; Chan et al., 2006), in part due to the long-recognized challenge of quantitatively assessing the aesthetic

value of landscape features (Leopold, 1969).

The links between ecosystem structure, function, and CES are often uncertain, which can make CES more difficult to articulate and quantify than other ecosystem services (Gee and Burkhard, 2010; Milcu et al., 2013; Norton et al., 2012; Plieninger et al., 2013). The ecosystem cascade model, where ecosystem services cascade directly from ecosystem structure and function (La Notte et al., 2017), is less likely to apply to CES because of the important role of human perceptions, culture, and demand in determining CES provisioning (Ernstson, 2013; Norton et al., 2012; Plieninger et al., 2013). Although other ecosystem services can be quantified using remotely sensed data, spatially-explicit estimates of CES across landscapes have had to rely on proxies, such as land cover or the presence of recreation facilities (Raudsepp-Hearne et al., 2010), that may not accurately represent CES.

In response to this need, methods using social media data, especially

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from photo-sharing sites such as Flickr, Panoramio, and Instagram, to estimate CES provisioning have proliferated in recent years (Figuerola-Alfaro and Tang, 2017; Ilieva and McPhearson, 2018; van Berkel et al., 2018). Since these data are often geolocated and the density of social media use is well correlated to visitation rates, social media data can be used to quantify spatial patterns of CES provisioning (Hamstead et al., 2018; Sonter et al., 2016; van Berkel et al., 2018; Wood et al., 2013). Social media has been used to understand how CES provisioning is related to landscape features, such as water, mountainous terrain, conserved land (Oteros-Rozas et al., 2018; Schirpke et al., 2018; Sonter et al., 2016; van Zanten et al., 2016), attributes of access such as roads, trails, and campgrounds (Ridding et al., 2018; Schirpke et al., 2018; Sonter et al., 2016; Walden-Schreiner et al., 2018), and demand, as indicated by proximity to urban centers (Ridding et al., 2018; Schirpke et al., 2018; van Zanten et al., 2016).

In addition to geolocation, photos shared on photo-sharing sites are associated with a variety of content. The image itself and associated text—captions, descriptions, tags—can be used to understand the types of CES provided over space. A small number of studies have used content analysis of images to understand variation across multiple CES (Hausmann et al., 2018; Heikinheimo et al., 2017; Martínez Pastur et al., 2016; van Berkel et al., 2018). These studies manually group images into a small number of *a priori* categories of CES based on visual content (e.g., recreation, aesthetics; Martínez Pastur et al., 2016; Van Berkel et al., 2018). These studies have found that different CES categories have distinct spatial patterns and relationships with landscape variables (Martínez Pastur et al., 2016; van Berkel et al., 2018).

However, visual content analysis of images only provides limited information and relies on a subjective classification with *a priori* categories. While useful information can be gleaned from visual analyses of the images themselves, this manual approach of coding images does not necessarily capture user intent and may restrict the depth and breadth of CES categories included. Indeed, CES research often focuses too narrowly on recreation and ecotourism services, which are the easiest to quantify and may have distinct spatial patterns and relationships with landscape features (Milcu et al., 2013). To address the limitations of visual content analysis, we developed a method to examine the content of the text (title, tags, and description) associated with Flickr images to quantify CES at landscape scales. By examining the social media users' words, we apply a bottom-up, data-driven approach to capture rich information about CES provisioning. Analysis of tags and other text associated with Flickr images has been used previously in urban areas to investigate collective understandings of urban spaces (Dunkel, 2015; Hollenstein and Purves, 2010). In CES studies, tags and comments associated with shared photographs have been used to qualitatively group images by landscape values (Chen et al., 2018) and tone (Barry, 2014), but no studies have taken a bottom-up, data-driven approach to quantifying CES based on social media text.

In this study, we used metadata associated with Flickr images to quantify CES associated with river ecosystems in Idaho. Rivers are popular recreational areas, and fishing and outdoor river recreation are major components of the cultural identity and economy of Idaho, USA. Aquatic features, such as rivers, are often identified as important sources of CES (Oteros-Rozas et al., 2018; van Zanten et al., 2016). Although research on ecosystem services derived from water is growing, the main focus has been on water quality, biodiversity, and agriculture, rather than on cultural ecosystem services (Aznar-Sánchez et al., 2019). We address four research questions: 1) What cultural ecosystem services do people obtain from rivers in Idaho? 2) How does overall CES provisioning vary across ecosystem characteristics? 3) How do specific CES relate to ecosystem characteristics? and 4) Do specific CES bundle in ways that parallel traditional CES categories? These research questions, and the methods applied here to address them, offer a rich understanding of how CES provisioning relates to ecosystem characteristics and an opportunity to improve landscape management and conservation decisions based on CES provisioning.

2. Methods:

2.1. Study sites: rivers of Idaho, USA

Idaho is a Western state in the USA with a rich set of ecological and social contexts; we focus on the ecosystems associated with 9734 river kilometers that run throughout these diverse landscapes. Idaho's rivers vary in size from small perennial streams to large river ecosystems, many of which are protected by state and federal conservation programs. Idaho's river ecosystems support 19 endangered fish species, eco-tourism, local economies, irrigation, and hydro-electric power. Land cover across the state is 52% scrubland, 32% forest, 11% agriculture, and 2% developed (Homer et al., 2015). Elevation in Idaho averages 1593 m and ranges from 216 to 3859 m above sea level. The dominant Köppen climate types in Idaho include humid continental (46%), dry continental (28%), and cold semi-arid (18%), while the higher elevation areas have a subarctic climate (5%); (PRISM Climate Group, 2004). Like many Western US states, a significant proportion of Idaho's land is owned and managed by government agencies (38% US Forest Service, 22% Bureau of Land Management, 5% other state and federal agencies), and a smaller proportion (30%) by private landowners. Tourism in Idaho is a US\$3.7 billion industry, attracting 34.3 million visitors annually, and visitation to parks, hiking and backpacking, and camping are among the top tourist activities in Idaho (Idaho Department of Commerce, 2018).

2.2. Flickr data

We used geo-referenced and tagged Flickr images as a proxy for visitation, preferences, and perceptions (Barry, 2014; Dunkel, 2015; van Zanten et al., 2016). Flickr data are more accessible than other sites, such as Facebook and Instagram, which have recently restricted data access. While there is variation of social media use across platforms, recent research shows overall spatial patterns are comparable among platforms (van Zanten et al., 2016) and Flickr represents a broader demographic than other social media sites (Cox et al., 2008). Rather than estimating the *potential* to provide CES, which could be characterized by an ecosystem without access, the method developed here quantifies the flow of CES and requires a person to receive the services.

Data were obtained from the Yahoo Flickr Creative Commons 100 M dataset, which includes the metadata for 100 million Flickr images across the world. Flickr metadata includes user name, image title, description, user tags, machine tags, and coordinates of geotagged images. For this analysis, we only used images with both a geotagged location and some text associated with description, title, or user tags. All text associated with the description, title, and user tags for each image was combined and analyzed as one data point.

For the image dataset, we selected images from within a 200-m buffer around rivers in Idaho. Using the National Hydrography Dataset (NHD), we created a river dataset that included all Idaho segments that were named "River" and excluded sections of rivers that were reservoirs. This dataset therefore excluded smaller streams and creeks. We manually added sections of rivers that were missing from the NHD using 2015 1-m color infrared orthorectified aerial images from the USDA National Agriculture Imagery Program (NAIP) to assess gaps in the NHD rivers (e.g., river sections were added if aerial photographs indicated that a river was present through a gap). The 200-m buffer was chosen to maximize sample size while maintaining high probability that pictures taken of, in, or near the river relate to CES provided by the river ecosystem. We chose this distance through an iterative process of selecting images within different buffers and assessing the frequency of the word *river* in the resulting dataset.

There were 6109 images taken within 200 m of a river between the years 2006, when geotagging functionality was added to Flickr, and 2014. To avoid missing data or double counting, we refined the dataset

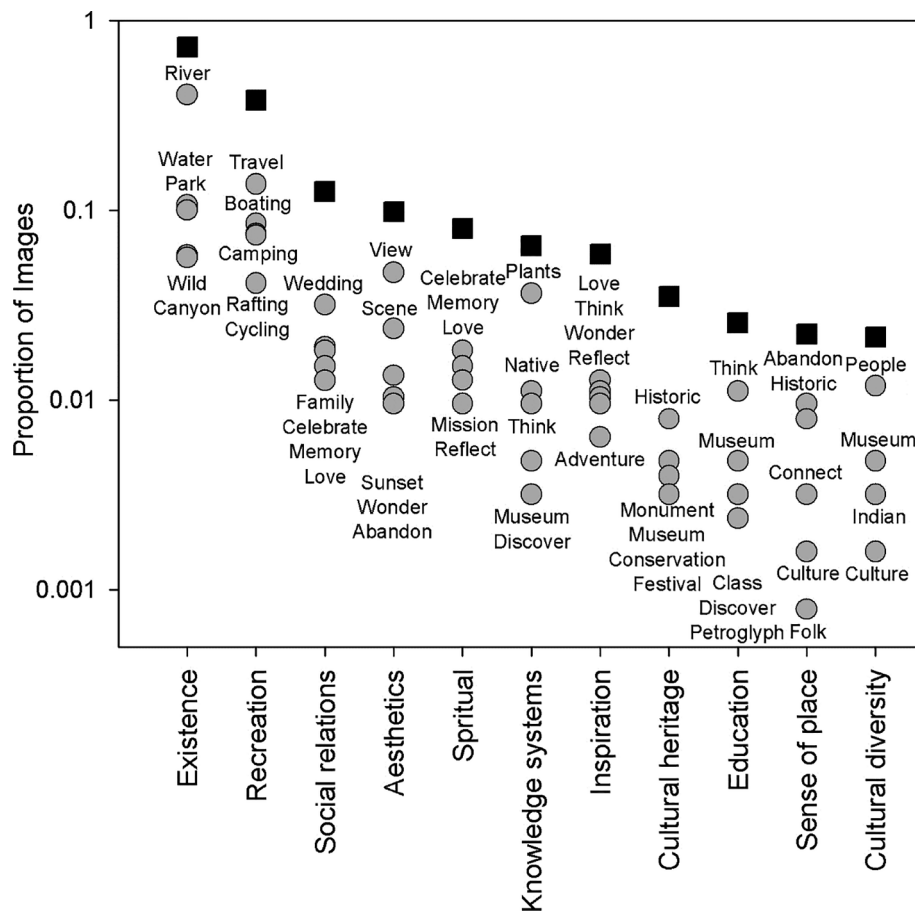


Fig. 1. Proportion of total images ($N = 1197$) with words in each CES category (black squares) and top 5 most frequent words from each CES category plotted by frequency of occurrence (grey circles). Note that an image could have words in multiple categories and so proportions can sum to > 1 .

by removing images without text and removing images with duplicate user, geolocation, and/or duplicate text (i.e., multiple images posted by the same user with identical text). Finally, we identified “active users” as a user with more than 50 images within the dataset and randomly selected 50 images for each user to be retained in the dataset. After this data cleaning process, there were 1197 images associated with the rivers of Idaho remaining for analysis.

2.3. Identifying CES provisioning from text analysis of geolocated Flickr images

To address our first question, what CES people obtain from rivers in Idaho, we used 11 previously-identified CES categories (e.g., recreation, aesthetics, etc.; Fig. 1) from the literature and key words or word groups (e.g., *river*, *boat*, *beautiful*) associated with each of those categories (Fig. 1; Milcu et al., 2013; Pleninger et al., 2013). Second, we cross-checked the word list from the literature review and added any additional words from the geospatial Flickr dataset that fit into one or more than one CES category. We then analyzed and coded the text from our geospatial Flickr dataset for the presence/absence of each word group. We did not include words that were common in our dataset and likely to refer to proper names (e.g., *snake*, *bear*, *church*, *salmon* are likely to be references to Snake River, Bear Lake, Frank Church Wilderness, or the Salmon River, respectively). Although some of these occurrences may refer to actual animals, we excluded all instances to be consistent and avoid introducing bias into our sample. To determine this list of excluded terms, we identified words in our dataset that were both very common and associated with a proper name. We mapped these words to see if they clustered in those areas with a proper name. We were only concerned with words that were very abundant and that

matched proper names of rivers and lakes that would fall within our sampling area. Words that were conceptually similar were combined into groups (e.g., *river*, *stream*, and *creek* were all coded as *river*). Although variations of similar words such as *river* and *creek* may carry different cultural significance (e.g., Lacy et al., 2018), combining similar terms allowed us to focus on broad patterns of features and CES. Based on the spatial location of sampled images, we assume that our data is representative of rivers only—not smaller streams. Word groups could be included in more than one CES category. Some CES categories were more difficult to operationalize, such as sense of place and cultural diversity, and had fewer word groups (Fig. 1). We found no instances of non-English words in this dataset.

2.4. Assessing variation in CES provisioning across ecosystem (social, biophysical, and infrastructure) characteristics

To assess how CES varied across social, biophysical, and infrastructure characteristics of river ecosystems, we used river reaches as our unit of analysis. Using ArcGIS, we divided the rivers of Idaho into 991 reaches approximately 10-km long, a size which allowed us to capture the heterogeneous social and biophysical context of CES provisioning. Reaches never spanned two rivers, but we did not consider other features, such as confluences, in making divisions within a single river. Because rivers range in length, some reaches were less than 10 km. The average reach length was 9.8 km. For each reach, we aggregated Flickr image data to calculate: presence or absence of images (as a proxy for CES provisioning), density of images (images/km of river), proportion of images with each coded word group, and presence or absence of each coded word group for each reach. Finally, we characterized the social, ecological, and built context of each river

Table 1
Social-ecological-built characteristics of 10-km river reaches in Idaho.

Variable Category	Variable	Value (standard deviation) across all reaches (N = 991)	
Access	Access Yes! Properties ¹	3.7	
(% reaches with access)	Campground ²	17.4	
	Birding Hotspot ³	5.5	
	Fishing or Boating Access ²	11.2	
Ownership ⁴	Bureau of Land Management	10.3	(22.8)
(Mean % ownership)	Private	37.0	(37.4)
	State	3.1	(10.4)
	U.S. Forest Service	35.5	(43.2)
	Tribal Lands	0.7	(6.3)
	National Park Service	0.004	(0.7)
	Other Federal Ownership	13.4	(17.5)
Land Cover ⁵	Developed	4.4	(8.5)
(Mean % cover)	Forest	31.2	(31.3)
	Crop	8.9	(18.8)
	Cliff	1.5	(5.8)
	Mining	0.1	(0.5)
	Pasture	3.3	(8.6)
	Scrub/Grass	25.8	(22.3)
	Water	7.8	(15.0)
	Wetland	15.5	(16.8)
Proximity (Mean distance)	Distance to urban area (km) ⁶	54.3	(38.5)
	Distance to road (km) ⁷	11.2	(15.1)
Biophysical	Elevation (m) ⁸	1447.8	(494.1)
	Relief (m)	144.0	(157.5)
	Annual precipitation (mm) ⁸	618	(313)
	Max temperature (°C) ⁸	13.4	(2.5)
	Impaired (% of streams) ⁹	0.3	
	Images / km	0.13	(0.71)

¹ Private property with public fishing and hunting access; Idaho Fish and Game.

² Idaho Parks and Recreation.

³ Idaho Fish and Game.

⁴ Surface Management Agency.

⁵ North West Regional Gap Analysis Project.

⁶ Calculated from US Census 2015 Urban Areas.

⁷ Calculated from US Census TigerLine.

⁸ PRISM.

⁹ EPA 303d list.

reach using a variety of publicly available datasets, including land ownership (social characteristics), land cover, vegetation type (ecological characteristics), public access, and proximity to roads and urban areas (characteristics related to built infrastructure; Table 1).

To answer question 2, “How does overall CES provisioning vary across ecosystem characteristics?” we first used logistic regression to model the probability of image presence (as a proxy for CES provisioning) along river reaches with different social, ecological, and built characteristics. Second, for reaches that had images, we used multiple linear regression to model the density of images in reaches of different social, ecological, and built characteristics. Image density was log transformed to improve data normality. All analyses were conducted in R (R Core Team, 2018). We used the package “glmulti” (Calcagno, 2013) for logistic regression model selection and AIC as model selection criteria. We used the package “leaps” (Lumley, 2017) for linear regression model selection and R² as the selection criteria.

2.5. Assessing variation in specific CES provisioning across ecosystem characteristics

To assess how specific CES provisioning was related to ecosystem characteristics (question 3), we used a redundancy analysis (RDA) to link specific words with social, ecological, and built characteristics of river reaches. RDA can be considered as an expansion of multiple linear regression, in that multiple explanatory variables are used to explain multiple response variables (Legendre and Legendre, 2012). We used only words that were present in at least 30 reaches (N = 19) to avoid rare words driving results. Ecosystem variables were same as above, with collinear variables (precipitation, maximum temperature, and forest land cover) removed to minimize collinearity, as assessed with variable inflation factor (VIF max < 10). We also removed variables for which there was limited data (i.e., if a variable was present in fewer than 10 reaches). Total number of river reaches included in this analysis was 222 (22% of all reaches).

Redundancy analysis models were run for both words and CES categories using the “vegan” package in R (Oksanen et al., 2018). We used backwards and forwards stepwise model selection based on R² to select the most parsimonious model. Model and axis significance were determined using a permutation analysis of variance with 1000 iterations.

2.6. Comparing CES bundles with traditional CES categories

Finally, to understand how specific CES bundle together, we used a hierarchical cluster analysis of words based on the first three axes of the RDA. We used the Ward method on a Euclidean distance matrix to identify clusters.

3. Results

3.1. What cultural ecosystem services do people obtain from rivers in Idaho?

The text associated with Flickr images provided rich source of information about CES. We identified 167 words linked to 11 CES categories in our dataset (Supplemental Material). The most commonly identified CES categories were Existence (72% of coded images), Recreation (38%), Social Relations (13%), and Aesthetics (10%; Fig. 1). Other CES categories were identified in < 10% of coded images (Fig. 1). Specific words provided important details about the nature of each of the broad CES categories. The 10 most frequently observed words were *river* (41% of images), *travel* (14%), *water* (11%), *park* (10%), *boating* (9%), *camping* (8%), *raft* (7%), *wild* (6%), *canyon* (6%), and *sun* (6%; Fig. 1, Supplemental Material).

3.2. How does overall CES provisioning vary across ecosystem characteristics?

CES provisioning, as reflected in the presence and density of images, was related to access to rivers, land cover, and land ownership. Of the 991 10-km river reaches in Idaho, 252 (~25%) had at least one image, a proxy for CES provisioning. Image densities for reaches with at least one image ranged from 0.1 to 17.6 images/river km, with a mean of 0.52 and a median of 0.20 images/river km (Fig. 2).

Access, land ownership, and land cover explained 21% of the variation in the presence of images, or CES provisioning, across river reaches in the best-fit logistic regression model (Table 2). Developed land cover had the strongest effect (Table 2); logistic regression with developed land cover alone explained 11% of the variation in image presence. Image presence was related to access to rivers through campgrounds and five land-ownership variables. Specifically, State Park ownership was positively associated with the presence of images in a river reach, while reaches with larger proportions in Tribal Lands, Bureau of Land Management (BLM), private, and US Forest Service ownership were less likely to have images (Table 2). Agricultural land

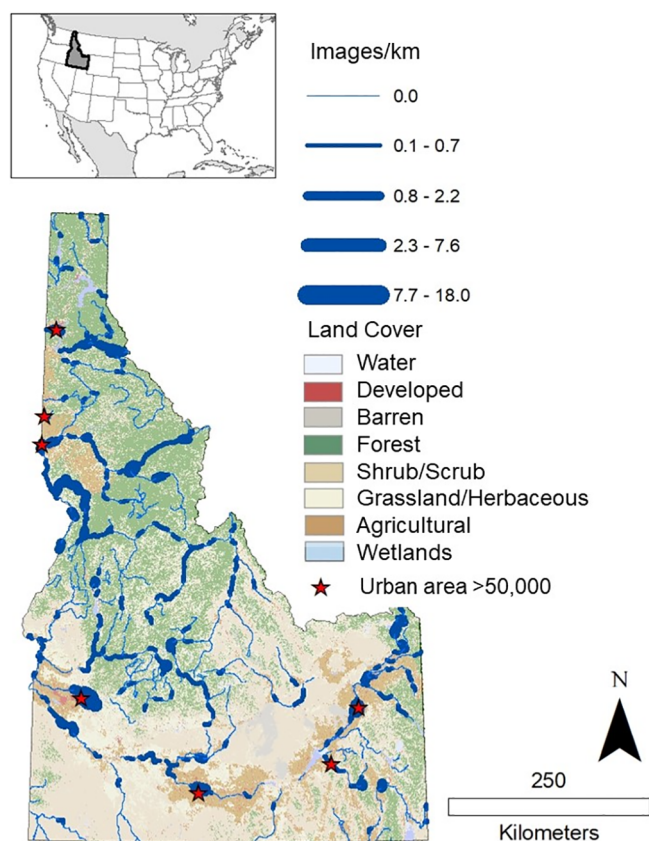


Fig. 2. Density of images (images/km) by reach across Idaho rivers. Mean density is 0.52 images/river km and median density is 0.20 images/river km. Land cover data from the National Land Cover Database (Homer et al., 2015).

cover—both crops and pasture—was negatively associated with images, as was forest land cover, while water cover was positively associated with images (Table 2). Relief was positively associated with image presence (Table 2).

Image densities (images/river km) were associated with a different set of landscape variables than image presence, although some variables were common to both models. The best-fit linear regression model explained 24% of the variation in image density (Table 2). Densities were more strongly related to land cover and ownership, rather than attributes of access. Developed land cover was the variable most strongly associated with image density (Table 2). Cliff land cover, State Park ownership, water cover, and elevation were all positively associated with higher densities of images (Table 2).

3.3. How do specific CES relate to ecosystem characteristics?

Specific cultural ecosystem services, as measured by words and CES categories, were significantly related to landscape variables, but the relationships were stronger for words than for categories. Landscape variables describing ownership, cover, and access determined the characteristics of CES provisioning, as reflected in words (Fig. 3). The best-fit RDA of words explained 14% of the variation. Both the overall model and the first three RDA axes were significant ($p < 0.01$). The first RDA axis was driven by a gradient from more remote (distance to the nearest road) areas under US Forest Service ownership to areas under private ownership with more developed land cover (Table 3, Fig. 3). The second RDA axis was most strongly associated with developed and scrub-grass land cover (Table 3). Remote areas, reflected by distance to the nearest road, were associated with the words *boat*, *wild*, *river*, and *camp*. US Forest Service ownership was strongly associated with recreation. Developed land cover and private ownership

Table 2

Social, biophysical and built landscape characteristics explaining the presence and density of CES provisioning (estimated with image presence and density). Results from best-fit logistic (presence or absence of images) and linear (density of images) regression models of overall CES provisioning. Coefficients for linear regression model have been back-transformed., significant coefficients are in bold. R^2 for logistic model represents McFadden's pseudo- R^2 .

Variable Category	Variable	Logistic Model Coefficient (Standard error)	Linear Model Coefficient (Standard error)	
Access	(Intercept)	−0.18 (0.39)	−0.13 (0.11)	***
	Campground	0.48 (0.22)		*
	Birding Hotspot		0.13 (0.07)	.
	Fishing or Boating Access		−0.08 (0.06)	
Ownership	State Park	6.09 (3.70)	1.04 (0.46)	.
	Tribal Lands	−3.36 (1.25)		*
	Bureau of Land Management	−2.59 (0.65)		***
	Private	−1.46 (0.44)		***
	U.S. Forest Service	−0.92 (0.42)	0.10 (0.06)	*
Land Cover	Developed	10.24 (1.21)	2.99 (0.20)	***
	Cliff	2.98 (1.86)	1.81 (0.54)	*
	Water	1.43 (0.61)	0.77 (0.15)	***
	Pasture	−3.40 (1.38)		*
	Crop	−1.76 (0.66)		**
	Forest	−1.56 (0.48)		**
	Scrub/Grass		0.13 (0.11)	
Biophysical	Elevation (m)		0.0001 (0.00005)	**
	Relief (m)	0.001 (0.0005)		*
Model Fit	N	976	241	
	R^2	0.21	0.24	
	P	< 0.0001	< 0.0001	

.p < 0.1, *p < 0.05, **p < 0.001, ***p < 0.0001.

were strongly associated with the words *cycle* and *road*. Cliff land cover was also retained in this model but was not strongly associated with any axes (Table 3, Fig. 3).

CES categories were also significantly related to landscape factors, but the RDA explained much less variation in CES categories (6%) than variation for words (Table 4; Fig. 4). CES categories were related most strongly to ownership, but also land cover and remoteness. The best-fit RDA and the first two RDA axes were significant ($p < 0.05$). Land ownership variables dominated the CES category RDA, including privately owned and BLM land. RDA axis one described a gradient from private land ownership to remote areas far from roads, and axis two described a gradient from wetland land cover to BLM ownership (Fig. 4, Table 4). Overall, recreation was provided in different landscapes compared to other CES categories; recreation was primarily associated with distance from road and BLM land.

3.4. Do specific CES bundle in ways that parallel traditional CES categories?

Finally, we found that specific CES, as reflected by words, bundled in ways that were distinct from traditional CES categories (Fig. 5). We identified four distinct CES clusters based on the hierarchical cluster analysis of words using scores from RDA axes 1–3. These clusters were distinct from traditional CES categories. Two (clusters 1 and 4) reflected specific activities: one included the terms *river*, *boat*, *wild*, *boat*, and *recreation*, suggesting wilderness boating, a subset of recreation. The second included the words *travel*, *cycle*, and *road* and suggests a mode of interaction—viewing rivers from the road—rather than a specific CES and draws from multiple traditional CES categories. These two clusters suggest very specific activities associated with rivers, and

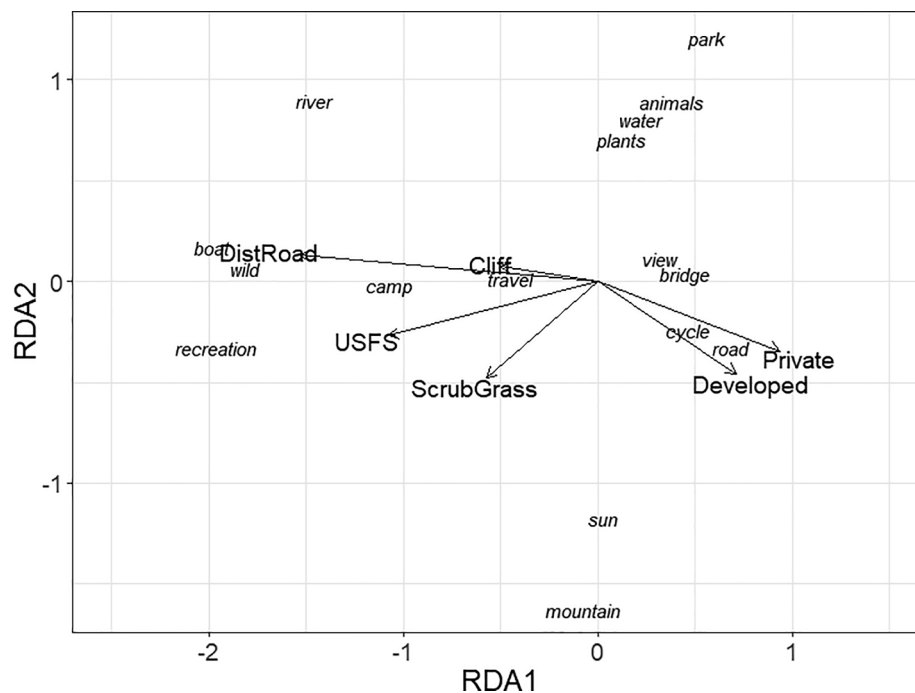


Fig. 3. Redundancy analysis biplot for CES words and their relationships to landscape variables. CES words (response variables) are smaller italicized font, landscape variables (predictor variables) are larger bold font.

Table 3

Redundancy analysis axes to explain variation in specific CES represented by words.

	RDA1	RDA2	RDA3
<i>P</i> (permutation test significance)	0.001	0.002	0.003
Eigenvalue	1.17	0.33	0.25
% Variance Explained	58.3	16.5	12.6
Cumulative % Variance Explained	58.3	74.7	87.4
Constraining Variable Contributions			
Cliff land cover	−0.31	0.06	−0.08
Distance to Road	−0.95	0.11	0.08
Private ownership	0.58	−0.29	0.57
US Forest Service	−0.67	−0.23	−0.02
Scrub-Grass land cover	−0.35	−0.40	−0.35
Developed land cover	0.44	−0.39	−0.42

the words include both activities (*camp*, *cycle*, *boat*) and specific landscape features (*river*, *road*). The remaining two clusters reflected specific aspects of either or both existence and aesthetic values and also included specific landscape features. One included *plants*, *animals*, *water*, *view*, and *bridge*, and the other included *mountain* and *sun* (Fig. 5).

4. Discussion

Quantifying CES at the landscape scale is increasingly recognized as key to conservation management (BenDor et al., 2017; Guerry et al., 2015; Mace, 2014; Pleasant et al., 2014; Plieninger et al., 2015). Thus, there is a need for decision-support tools and approaches that can incorporate CES into landscape-scale conservation decision-making (Guerry et al., 2015). Yet methods for quantifying CES at the landscape scale have lagged those for provisioning, supporting, and regulating services (Casalegno et al., 2013; Chan et al., 2006). We found that the text associated with social media images, such as Flickr, can be used to identify a wide range of CES, including those that are not typically examined in empirical studies (e.g., Milcu et al., 2013; Fig. 1). Further, we used image-associated text to quantify relationships between landscape structure and CES provisioning and explore associations among

Table 4

Redundancy analysis axes to explain variation in CES categories.

	RDA1	RDA2
<i>P</i> (permutation test significance)	0.002	0.031
Eigenvalue	0.38	0.17
% Variance Explained	53.9	25.0
Cumulative % Variance Explained	53.9	78.9
Constraining Variable Contributions		
Private ownership	−0.79	−0.14
Distance to Road	0.83	0.23
Bureau of Land Management	−0.07	0.76
Wetland Land cover	−0.17	−0.64

specific CES (Figs. 2–4). Overall CES provisioning, as well as provisioning of specific CES, was significantly but weakly associated with access, land cover, and land ownership. Furthermore, landscape variables were more strongly associated with specific CES than general CES categories, and bundles of CES based on landscape associations were distinct from traditional CES categories. Text associated with social media images can be a powerful way to understand the nuances of CES provisioning, adding richness and context beyond traditional CES categories. The method developed here offers a valuable decision support tool to conservation and funding agencies interested in managing and preserving CES for current and future generations.

4.1. Links between ecological, social, and built landscape structure and CES

Overall, we found evidence that CES provisioning is significantly, though weakly, related to the ecological aspects of the landscape. Our findings highlight that the cascade model, in which ecosystem services flow from ecosystem structure (La Notte et al., 2017), has limited utility for specifically addressing CES. Given the importance of culture, identity, and individual preferences in determining CES (Norton et al., 2012), the lack of explanatory power of models relating CES and ecosystem structure is not surprising. The results highlight the critical challenges in quantifying CES across landscape scales, a task which requires strong predictive relationships between landscape structure

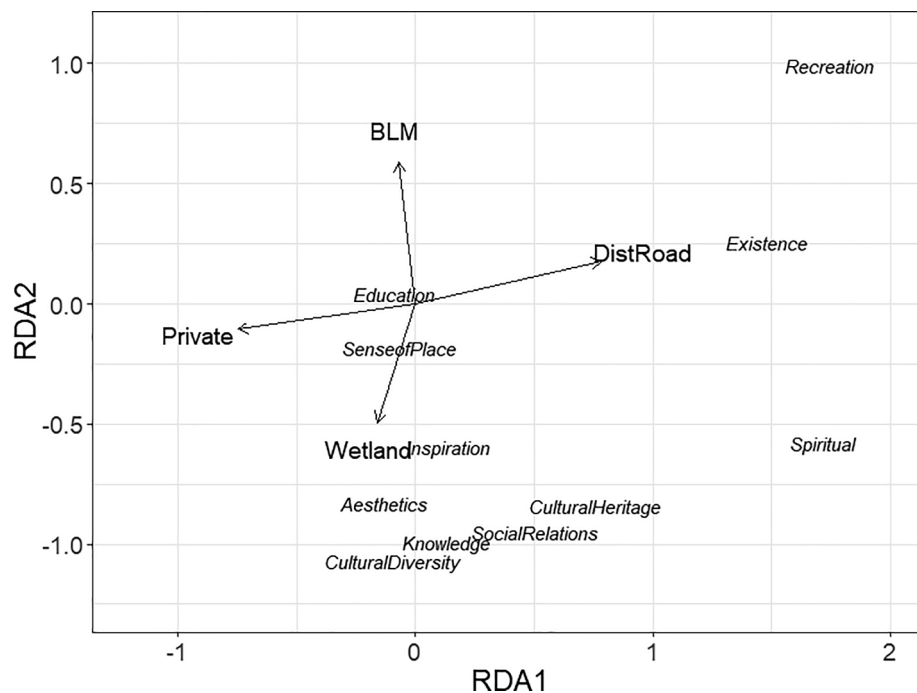


Fig. 4. Redundancy analysis biplot of CES categories and their relationships to landscape variables. CES words (response variables) are smaller italicized font, landscape variables (predictor variables) are larger bold font.

and CES provisioning (Burkhard et al., 2012; Casalegno et al., 2013). Including social and built aspects of landscape structure increased our understanding of CES beyond the traditional cascade model, but still left most (i.e., greater than 86%) of the variation in provisioning unexplained. The social and cultural attributes of CES beneficiaries is likely important in explaining the remaining variation and CES demand (Bennett et al., 2009; Cord et al., 2017; Plieninger et al., 2013).

Despite explaining less variation than other landscape features, ecological characteristics, including land covers of water, forest, developed, and cliff, were significantly associated with CES provisioning as a whole and specific individual CES (Tables 2 and 4). For example, river reaches had higher image density, and thus provided more CES, when they had more water and less forest land cover. This finding suggests that larger rivers are more likely to provide CES than smaller rivers, perhaps because larger rivers are likely to be more accessible or have greater

cultural significance (Lacy et al., 2018). Additionally, our findings support several earlier studies that have found negative relationships between forest cover and CES provisioning, possibly due to trees impeding landscape views (Sonter et al., 2016; van Berkel et al., 2018). In semi-arid Idaho, forest land cover is more common at higher, less accessible elevations, or the relationship could reflect the negative correlation between forest and developed land, which was strongly and positively associated with image presence and density. Cliff land cover was associated with higher densities of CES generally and wilderness recreation specifically. This suggests the importance of visually striking landscape features in providing CES, supporting earlier findings that topography and viewshed area were positively associated with CES (Ridding et al., 2018; van Berkel et al., 2018; Walden-Schreiner et al., 2018).

We also found that while CES provisioning was related to landscape structure, social and built elements of the landscape were as or more

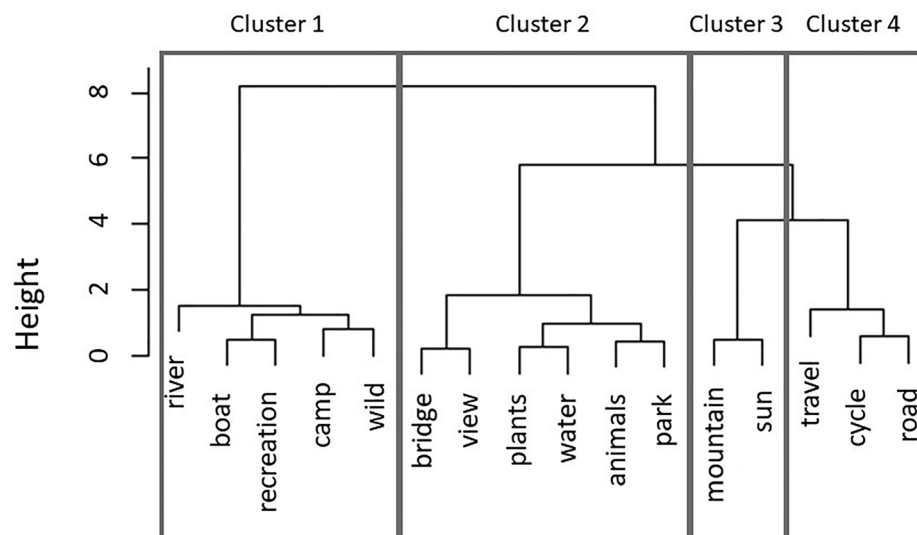


Fig. 5. Specific CES, as reflected by words, bundle into distinct clusters. Cluster 1 and 4 identify specific recreational activities. Clusters 2 and 3 reflect existence or aesthetic values, and all clusters include words describing specific landscape features (e.g., river, bridge, water, park, mountain, road).

important than ecological elements. This suggests a critical distinction between CESs and other ecosystem services: while other ecosystem services, such as food and fiber provisioning, may be supported by intermediaries to enhance the delivery of the services from source ecosystems to beneficiaries, CES beneficiaries *require* intermediaries to derive CES, including physical access to ecosystems via proximity (van Zanten et al., 2016) and physical infrastructure such as roads, access points, and maintained facilities (Richards and Friess, 2015; Walden-Schreiner et al., 2018).

Both supply and demand determine the flow of all ecosystem services (Burkhard et al., 2012). Supply of CES may be hindered, for example, by degraded ecosystems or a lack of access to particular ecosystems, but demand is especially important for determining flows of CES. The demand for CES can be affected by proximity to higher density populations, where a greater number of people are likely to be in demand of services, and cultural factors, such as the historical or religious importance of an ecosystem (Cord et al., 2017; Keeler et al., 2015; Richards and Friess, 2015; Ridding et al., 2018; Schirpke et al., 2018; Sonter et al., 2016; Walden-Schreiner et al., 2018). Our analysis reflects the importance of demand through the strong effect of developed land cover, which is related to population density, in driving CES provisioning. Areas with high population density are likely to provide greater flows of CES simply by virtue of having a large number of proximate beneficiaries. Publicly available data, along with the content analysis method presented here for estimating CES from social media allows us to easily capture these important drivers of CES.

Social dimensions of the landscape, such as land ownership, affect access by visitors and the provision of CES. We found that land ownership and land use were the strongest predictors of overall CES provisioning. CES provisioning was most strongly predicted by developed land (Table 2). Developed land influences both access and demand; proximity to population centers allows easy access and demand. Agricultural land for both pasture and crops was negatively associated with CES provisioning, as was private land, likely due to reduced access to these river reaches. Private and agricultural lands likely provide CES, particularly a sense of place and rural way of life, to land owners and residents (van Berkel and Verburg, 2014; van Zanten et al., 2016). On private agricultural lands, CES may not be captured by social media datasets because the beneficiaries might be less likely to share these experiences online. CES was the most strongly and positively associated with State Park ownership. US Forest Service land was also significantly associated with recreation in general and wilderness boating (Fig. 3), while BLM land was associated with the CES category recreation (Fig. 4).

Overall, the results from multiple analyses suggest recreational uses are primarily related to land ownership (Fig. 3), while other CES categories are related to other biophysical and built characteristics of the landscape. For example, CES related to spiritual fulfillment, cultural heritage, and aesthetic appreciation were more related to land cover and distance to the road or developed space (Fig. 4). Although access is required to derive CES benefits, distance to road was not significantly associated with overall CES provisioning, as measured by image density. This finding contrasts with other studies that found trail and road access to be important predictors of overall CES provisioning (Schirpke et al., 2018; Sonter et al., 2016; Walden-Schreiner et al., 2018). Although not important for overall CES provisioning, distance from road emerged as the most important factor distinguishing among specific CES and traditional CES categories. More remote areas were associated with existence, spiritual, and recreation values, especially wilderness boating, values. In contrast, the word *road* was common in our dataset and was associated with developed land cover and private ownership, highlighting that CES benefits can be derived from adjacent systems. That we found different landscape associations between specific and overall CES provisioning highlights an important implication of our analysis – if we had only used image density as our measure of CES (as is often done in the literature), we would have missed the importance of road access as a driver of CES provisioning.

4.2. Value of text analysis for understanding specific CES provisioning

Previous studies have used social media data to assess CES provisioning based on geo-tagged location of shared images (Sonter et al., 2016; Tenerelli et al., 2016; van Berkel et al., 2018; Walden-Schreiner et al., 2018; Yoshimura and Hiura, 2017) or labor intensive visual content analysis (Hausmann et al., 2018; Heikinheimo et al., 2017; Martínez Pastur et al., 2016; van Berkel et al., 2018). In addition, these past studies primarily examine only the dominant CES categories, such as recreation and aesthetics. The method presented here integrates a quantitative content analysis of text associated with shared images and adds richness to understanding CES provisioning at landscape scales. Not limited to a small number of categories, we identified 167 words that reflect specific CES: e.g., specific types of recreation (e.g., *cycling*, *boating*) or existence values associated with specific landscape or ecological elements (e.g., *rivers*, *mountains*, *plants*, *animals*; Fig. 1). This detailed analysis expands our ability to examine diverse, but understudied, CES and provides a more nuanced understanding of CES provided at large scales across diverse landscapes (Fig. 2). More thorough accounting of CES at landscape scales, like the one we present here, is critical to informing landscape-scale conservation efforts that aim to incorporate human benefits from the environment.

We found that specific CES as reflected by words were better predicted by landscape variables than traditional CES categories. This suggests that while traditional categories of CES are conceptually useful (e.g., recreation, existence, cultural heritage), they are not the best level of organization for understanding linkages between landscape factors and CES provisioning. Milcu et al. (2013) raised a concern that a focus on recreation and the assumption that recreation and ecotourism reflect all CES could marginalize other CES. In fact, our results suggest that recreational services are distinct from other CES in their spatial patterns and association with landscape factors. Traditional categories also fail to reflect how individual CES group over space: when we clustered words based on their associations with landscape variables, we found clusters that were distinct from traditional categories. It is intuitive that more specific CES (whitewater boating vs. recreation or appreciation of mountains vs. aesthetic or existence values) are more closely tied to landscape features. Text provides information-rich source for moving beyond a small number of broad CES categories and understanding not only the diversity of ways people interact with landscapes, but what landscape features support these uses. This understanding will be critical for comprehensive landscape-scale planning and conservation.

4.3. Limitations and future directions

While social media sources, such as Flickr, provide a cost-efficient method for crowdsourcing “big data” over large spatial and temporal scales, the limitations of CES quantification from social media data have been well described by others (Chen et al., 2018; Heikinheimo et al., 2017; Ilieva and McPhearson, 2018). Although social media is a convenient data source, it only represents the ideas and viewpoints of a subset of the population, and unfortunately, demographic data is often not available to evaluate the effect of social factors on shared content (Chen et al., 2018). Furthermore, a certain level of visitation and social media posts are needed to describe an area well (Chen et al., 2018; Heikinheimo et al., 2017). For example, areas with low use are not well represented in these datasets. Although low visitation rates and use offers important insight into the quantity of CES provided from that landscape, it does not address the quality or type of CES. Finally, some CES will not be as well represented as others through these data sources (e.g., inspiration, knowledge systems; Chen et al., 2018; Heikinheimo et al., 2017). Although some researchers have found good correlations between social media content and survey data (Hausmann et al., 2018; Heikinheimo et al., 2017), more work is needed to validate people's perceptions of the landscape, calibrate social media datasets with on-the-ground surveys, and quantify biases. Because social media data

include temporal as well as spatial metadata, these data can also be used to quantify the spatiotemporal dynamics of CES provisioning (Heikinheimo et al., 2017; Schirpke et al., 2018; Sonter et al., 2016), identifying overall temporal trends as well as spatiotemporal “hotspots” of CES provisioning throughout the year. Social media data have also been used to estimate residency of social media users based on common areas of uploads (e.g., Heikinheimo et al., 2017; Keeler et al., 2015). The effects of local residency on CES use has been explored within settings such as national parks, where results largely mirror those from visitor surveys (Heikinheimo et al., 2017), but people have also been observed to travel farther for high-quality ecosystems (e.g., lakes with cleaner water, Keeler et al., 2015). Future work could explore how these dynamics play out across ecosystems and land ownership. We might expect, for example, that urban rivers provide high flows of CES to proportionately more locals, while more remote and “wild” rivers provide CES to proportionately fewer visitors. These analyses would have important implications for the economics and management of CES across landscapes and scales.

5. Conclusion

Although social media data has limitations that need to be properly quantified, social media data provide a useful tool for quickly and inexpensively quantifying CES provisioning at regional scales. Text associated with images can be a powerful resource for understanding the nuances and specifics of CES provisioning and adding richness beyond traditional CES categories. Specific CES were more strongly associated with landscape characteristics than general CES categories (e.g., recreation, aesthetics). Importantly, a combination of social, built, and biophysical landscape features is required for CES provisioning. For the purposes of landscape planning, social media is an important tool for quantifying what CES are being provided and where in the landscape. These data need to be combined with both landscape information and socio-cultural data to understand the mechanisms underlying CES provisioning and anticipate future changes in CES provisioning at landscape scales.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2019.105580>.

References

- Aznar-Sánchez, J.A., Velasco-Muñoz, J.F., Belmonte-Ureña, L.J., Manzano-Agugliaro, F., 2019. The worldwide research trends on water ecosystem services. *Ecol. Ind.* 99, 310–323. <https://doi.org/10.1016/j.ecolind.2018.12.045>.
- Barry, S.J., 2014. Using social media to discover public values, interests, and perceptions about cattle grazing on park lands. *Environ. Manage.* 53, 454–464. <https://doi.org/10.1007/s00267-013-0216-4>.
- BenDor, T.K., Spurlock, D., Woodruff, S.C., Olander, L., 2017. A research agenda for ecosystem services in American environmental and land use planning. *Cities* 60, 260–271. <https://doi.org/10.1016/j.cities.2016.09.006>.
- Bennett, E.M., Peterson, G.D., Gordon, L.J., 2009. Understanding relationships among multiple ecosystem services. *Ecol. Lett.* 12, 1394–1404.
- Burkhard, B., Kroll, F., Nedkov, S., Müller, F., 2012. Mapping ecosystem service supply, demand and budgets. *Challenges of sustaining natural capital and ecosystem services Quantification, modelling & valuation/accounting. Ecol. Indic.* 21, 17–29. <https://doi.org/10.1016/j.ecolind.2011.06.019>.
- Calcagno, V., 2013. glmulti: Model selection and multimodel inference made easy. R package version 1.0.7. <https://CRAN.R-project.org/package=glmulti>.
- Casalegno, S., Inger, R., Desilvey, C., Gaston, K., 2013. Spatial covariance between aesthetic value & other ecosystem services. *PLoS ONE* 8, e68437. <https://doi.org/10.1371/journal.pone.0068437>.
- Chan, K.M.A., Shaw, M.R., Cameron, D.R., Underwood, E.C., Daily, G.C., 2006. Conservation planning for ecosystem services. *PLoS Biol.* 4, e379. <https://doi.org/10.1371/journal.pbio.0040379>.
- Chen, Y., Parkins, J.R., Sherren, K., 2018. Using geo-tagged Instagram posts to reveal landscape values around current and proposed hydroelectric dams and their reservoirs. *Landscape Urban Plann.* 170, 283–292. <https://doi.org/10.1016/j.landurbplan.2017.07.004>.
- Cord, A.F., Brauman, K.A., Chaplin-Kramer, R., Huth, A., Ziv, G., Seppelt, R., 2017. Priorities to advance monitoring of ecosystem services using Earth observation. *Trends Ecol. Evol.* 32, 416–428. <https://doi.org/10.1016/j.tree.2017.03.003>.
- Cox, A.M., Clough, P., Marlow, J., 2008. Flickr: a first look at user behaviour in the context of photography as serious leisure. *Inf. Res.* 13.
- Dunkel, A., 2015. Visualizing the perceived environment using crowdsourced photo geodata. Special Issue: Critical Approaches to Landscape Visualization. *Landscape Urban Plann.* 142, 173–186. <https://doi.org/10.1016/j.landurbplan.2015.02.022>.
- Ernstson, H., 2013. The social production of ecosystem services: a framework for studying environmental justice and ecological complexity in urbanized landscapes. Special Issue: Urban Ecosystem Services. *Landscape Urban Plann.* 109, 7–17. <https://doi.org/10.1016/j.landurbplan.2012.10.005>.
- Figueroa-Alfaro, R.W., Tang, Z., 2017. Evaluating the aesthetic value of cultural ecosystem services by mapping geo-tagged photographs from social media data on Panoramio and Flickr. *J. Environ. Plann. Manage.* 60, 266–281. <https://doi.org/10.1080/09640568.2016.1151772>.
- Gee, K., Burkhard, B., 2010. Cultural ecosystem services in the context of offshore wind farming: a case study from the west coast of Schleswig-Holstein. *Ecosystem Services – Bridging Ecology Economy and Social Sciences. Ecol. Complexity* 7, 349–358. <https://doi.org/10.1016/j.ecocom.2010.02.008>.
- Guerry, A.D., Polasky, S., Lubchenco, J., Chaplin-Kramer, R., Daily, G.C., Griffin, R., Ruckelshaus, M., Bateman, I.J., Duraiappah, A., Elmqvist, T., Feldman, M.W., Folke, C., Hoekstra, J., Kareiva, P.M., Keeler, B.L., Li, S., McKenzie, E., Ouyang, Z., Reyers, B., Ricketts, T.H., Rockström, J., Tallis, H., Vira, B., 2015. Natural capital and ecosystem services informing decisions: From promise to practice. *PNAS* 112, 7348–7355. <https://doi.org/10.1073/pnas.1503751112>.
- Hamstead, Z.A., Fisher, D., Ilieva, R.T., Wood, S.A., McPhearson, T., Kremer, P., 2018. Geolocated social media as a rapid indicator of park visitation and equitable park access. *Comput. Environ. Urban Syst.* 72, 38–50. <https://doi.org/10.1016/j.compenurbysys.2018.01.007>.
- Hausmann, A., Toivonen, T., Slotow, R., Tenkanen, H., Moilanen, A., Heikinheimo, V., Minin, E.D., 2018. Social media data can be used to understand tourists' preferences for nature-based experiences in protected areas. *Conserv. Lett.* 11, e12343. <https://doi.org/10.1111/conl.12343>.
- Heikinheimo, V., Minin, E.D., Tenkanen, H., Hausmann, A., Erkkonen, J., Toivonen, T., 2017. User-generated geographic information for visitor monitoring in a national park: a comparison of social media data and visitor survey. *ISPRS Int. J. Geo-Inf.* 6, 85. <https://doi.org/10.3390/ijgi6030085>.
- Hollenstein, L., Purves, R., 2010. Exploring place through user-generated content: using Flickr tags to describe city cores. *J. Spatial Inf. Sci.* 2010, 21–48. <https://doi.org/10.5311/JOSIS.2010.1.13>.
- Homer, C.G., Dewitz, J.A., Yang, L., Jin, S., Danielson, P., Xian, G., Coulston, J., Herold, N.D., Wickham, J.D., Megown, K., 2015. Completion of the 2011 National Land Cover Database for the conterminous United States-Representing a decade of land cover change information. *Photogramm. Eng. Remote Sens.* 81, 345–354.
- Idaho Department of Commerce, 2018. Tourism Research [WWW Document]. Idaho Commerce. URL <https://commerce.idaho.gov/tourism-resources/tourism-industry-development/research/> (accessed 11.19.18).
- Ilieva, R.T., McPhearson, T., 2018. Social-media data for urban sustainability. *Nat. Sustainability* 1, 553. <https://doi.org/10.1038/s41893-018-0153-6>.
- Keeler, B.L., Wood, S.A., Polasky, S., Kling, C., Filstrup, C.T., Downing, J.A., 2015. Recreational demand for clean water: evidence from geotagged photographs by visitors to lakes. *Front. Ecol. Environ.* 13, 76–81. <https://doi.org/10.1890/140124>.
- La Notte, A., D'Amato, D., Mäkinen, H., Paracchini, M.L., Liqueste, C., Ego, B., Geneletti, D., Crossman, N.D., 2017. Ecosystem services classification: a systems ecology perspective of the cascade framework. *Ecol. Ind.* 74, 392–402. <https://doi.org/10.1016/j.ecolind.2016.11.030>.
- Lacy, S.N., Di Giminiani, P., Mao, L., 2018. What defines a river? Modelling the interplay between physical and social driving factors in characterising the waterways in Chile. *Geog. J.* 184, 342–356. <https://doi.org/10.1111/geog.12262>.
- Legendre, P., Legendre, L.F.J., 2012. *Numerical Ecology*. Vol. 24. third ed. Elsevier, Amsterdam.
- Leopold, L.B., 1969. Quantitative comparison of some aesthetic factors among rivers (USGS Numbered Series No. 620). Circular. U.S. Geological Survey, Reston, VA.
- Lumley, T. based on Fortran code by Alan Miller. 2017. leaps: Regression Subset Selection. R package version 3.0. <https://CRAN.R-project.org/package=leaps>.
- Mace, G.M., 2014. Whose conservation? *Science* 345, 1558–1560. doi:10.1126/science.1254704.
- Martínez Pastur, G., Peri, P.L., Lencinas, M.V., García-Llorente, M., Martín-López, B., 2016. Spatial patterns of cultural ecosystem services provision in Southern Patagonia. *Landscape Ecol.* 31, 383–399. <https://doi.org/10.1007/s10980-015-0254-9>.
- Milcu, A.I., Hanspach, J., Abson, D., Fischer, J., 2013. Cultural ecosystem services: a literature review and prospects for future research. *Ecol. Soc.* 18.
- Millennium Ecosystem Assessment, 2005. *Ecosystems and Human Well-Being: Synthesis*. Island Press.
- Norton, L.R., Inwood, H., Crowe, A., Baker, A., 2012. Trialling a method to quantify the ‘cultural services’ of the English landscape using Countryside Survey data. *Land Use Policy* 29, 449–455. <https://doi.org/10.1016/j.landusepol.2011.09.002>.

- Oksanen, J., Blanchet F.G., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., Minchin, P. R., O'Hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Szoecs, E., Wagner, H., 2018. *vegan: Community Ecology Package*, package version 2.5-2. <https://CRAN.R-project.org/package=vegan>.
- Oteros-Rozas, E., Martín-López, B., Fagerholm, N., Bieling, C., Plieninger, T., 2018. Using social media photos to explore the relation between cultural ecosystem services and landscape features across five European sites. *Ecological Indicators Landscape Indicators – Monitoring of Biodiversity and Ecosystem Services at Landscape Level* 94, 74–86. <https://doi.org/10.1016/j.ecolind.2017.02.009>.
- Pleasant, M.M., Gray, S.A., Lepczyk, C., Fernandes, A., Hunter, N., Ford, D., 2014. Managing cultural ecosystem services. *Ecosyst. Serv.* 8, 141–147. <https://doi.org/10.1016/j.ecoser.2014.03.006>.
- Plieninger, T., Bieling, C., Fagerholm, N., Byg, A., Hartel, T., Hurley, P., López-Santiago, C.A., Nagabhatla, N., Oteros-Rozas, E., Raymond, C.M., van der Horst, D., Huntsinger, L., 2015. The role of cultural ecosystem services in landscape management and planning. *Curr. Opin. Environ. Sustain., Open Issue* 14, 28–33. <https://doi.org/10.1016/j.cosust.2015.02.006>.
- Plieninger, T., Dijk, S., Oteros-Rozas, E., Bieling, C., 2013. Assessing, mapping, and quantifying cultural ecosystem services at community level. *Land Use Policy* 33, 118–129. <https://doi.org/10.1016/j.landusepol.2012.12.013>.
- PRISM Climate Group, 2004. *PRISM. Oregon State University*.
- Core Team, R., 2018. *R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Australia*.
- Raudsepp-Hearne, C., Peterson, G.D., Bennett, E.M., 2010. Ecosystem service bundles for analyzing tradeoffs in diverse landscapes. *PNAS* 107, 5242–5247. <https://doi.org/10.1073/pnas.0907284107>.
- Richards, D.R., Friess, D.A., 2015. A rapid indicator of cultural ecosystem service usage at a fine spatial scale: content analysis of social media photographs. *Ecol. Ind.* 53, 187–195. <https://doi.org/10.1016/j.ecolind.2015.01.034>.
- Ridding, L.E., Redhead, J.W., Oliver, T.H., Schmucki, R., McGinlay, J., Graves, A.R., Morris, J., Bradbury, R.B., King, H., Bullock, J.M., 2018. The importance of landscape characteristics for the delivery of cultural ecosystem services. *J. Environ. Manage.* 206, 1145–1154. <https://doi.org/10.1016/j.jenvman.2017.11.066>.
- Schirpke, U., Meisch, C., Marsoner, T., Tappeiner, U., 2018. Revealing spatial and temporal patterns of outdoor recreation in the European Alps and their surroundings. *Assessment and Valuation of Recreational Ecosystem Services. Ecosyst. Serv.* 31, 336–350. <https://doi.org/10.1016/j.ecoser.2017.11.017>.
- Sonter, L.J., Watson, K.B., Wood, S.A., Rickerts, T.H., 2016. Spatial and temporal dynamics and value of nature-based recreation, estimated via social media. *PLOS ONE* 11, e0162372. <https://doi.org/10.1371/journal.pone.0162372>.
- Tenerelli, P., Demšar, U., Luque, S., 2016. Crowdsourcing indicators for cultural ecosystem services: a geographically weighted approach for mountain landscapes. *Ecol. Ind.* 64, 237–248. <https://doi.org/10.1016/j.ecolind.2015.12.042>.
- van Berkel, D.B., Tabrizian, P., Dörning, M.A., Smart, L., Newcomb, D., Mehaffey, M., Neale, A., Meentemeyer, R.K., 2018. Quantifying the visual-sensory landscape qualities that contribute to cultural ecosystem services using social media and LiDAR. *Ecosyst. Serv.* <https://doi.org/10.1016/j.ecoser.2018.03.022>.
- van Berkel, D.B., Verburg, P.H., 2014. Spatial quantification and valuation of cultural ecosystem services in an agricultural landscape. *Ecol. Ind.* 37, 163–174. <https://doi.org/10.1016/j.ecolind.2012.06.025>.
- van Zanten, B.T., van Berkel, D.B., Meentemeyer, R.K., Smith, J.W., Tieskens, K.F., Verburg, P.H., 2016. Continental-scale quantification of landscape values using social media data. *PNAS* 113, 12974–12979. <https://doi.org/10.1073/pnas.1614158113>.
- Walden-Schreiner, C., Leung, Y.-F., Tateosian, L., 2018. Digital footprints: incorporating crowdsourced geographic information for protected area management. *Appl. Geogr.* 90, 44–54. <https://doi.org/10.1016/j.apgeog.2017.11.004>.
- Wood, S.A., Guerry, A.D., Silver, J.M., Lacayo, M., 2013. Using social media to quantify nature-based tourism and recreation. *Sci. Rep.* 3, 2976. <https://doi.org/10.1038/srep02976>.
- Yoshimura, N., Hiura, T., 2017. Demand and supply of cultural ecosystem services: use of geotagged photos to map the aesthetic value of landscapes in Hokkaido. *Ecosyst. Serv.* 24, 68–78. <https://doi.org/10.1016/j.ecoser.2017.02.009>.